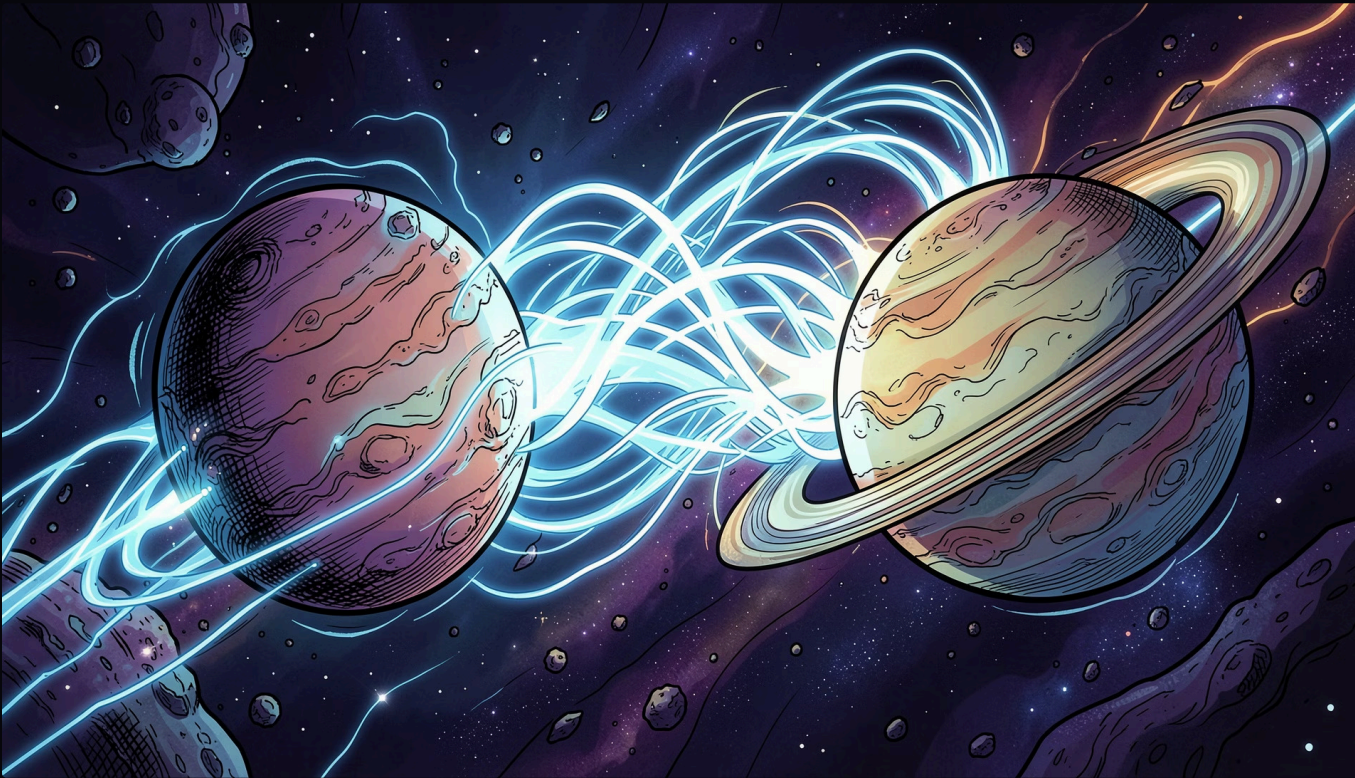


Chapter 8: Gravitation

Complete study notes covering Newton's Law of Gravitation, Kepler's Laws, acceleration due to gravity, gravitational potential energy, escape velocity, and satellites.



Newton's Law of Gravitation



Statement

Every particle in the universe attracts every other particle with a force directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

Formula

$$F = \frac{G \cdot m_1 \cdot m_2}{r^2}$$

- **G** = Universal Gravitational Constant = $6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
- **m₁, m₂** = masses of two bodies
- **r** = distance between centers
- Gravitational force is always **attractive** and acts along the line joining the centers.

Kepler's Laws of Planetary Motion

1

Law of Orbits (1st Law)

Each planet revolves around the Sun in an **elliptical orbit**, with the Sun at one of the two foci.

2

Law of Areas (2nd Law)

The radius vector from the Sun to a planet sweeps out **equal areas in equal intervals of time**. This is a consequence of conservation of angular momentum.

3

Law of Periods (3rd Law)

The square of the time period of revolution is directly proportional to the cube of the semi-major axis of the orbit.

$$T^2 \propto a^3$$

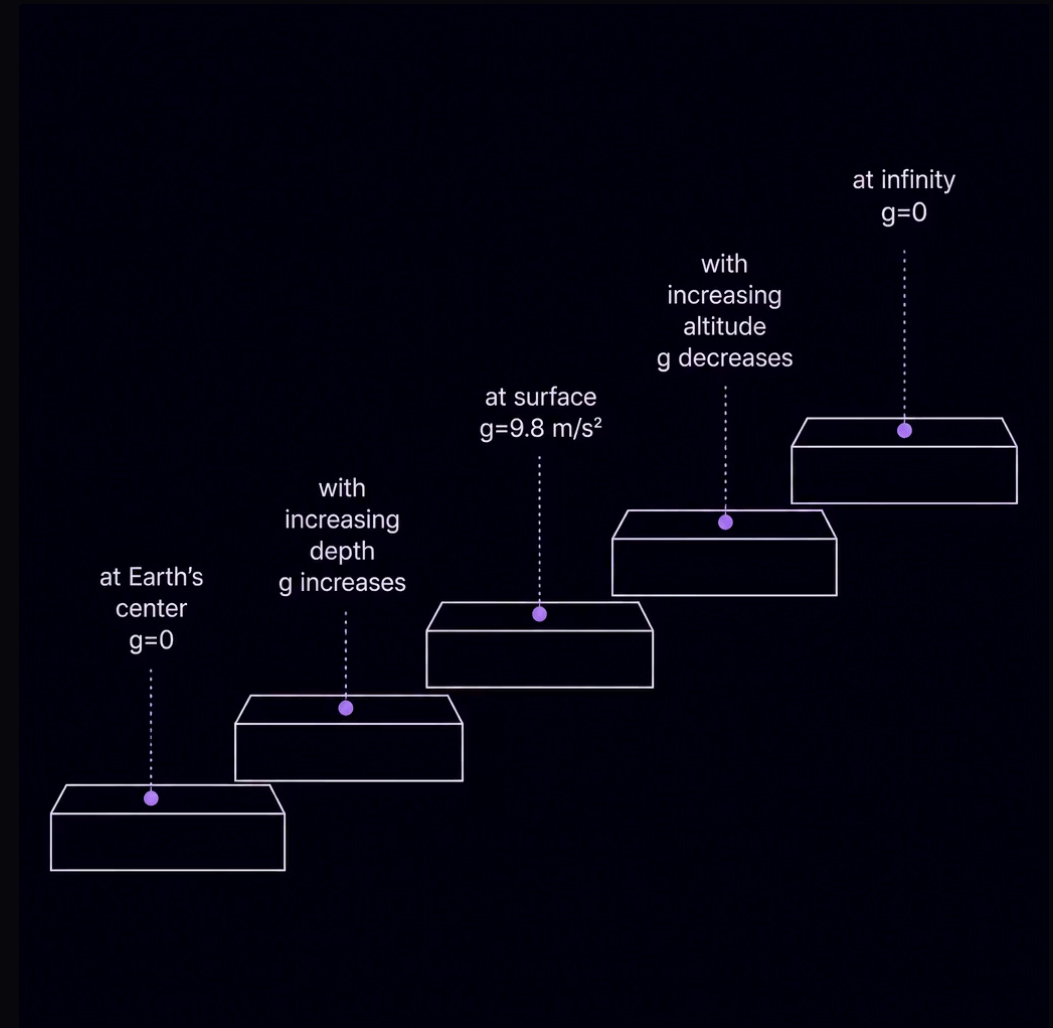
Acceleration Due to Gravity (g)

Formula on Earth's Surface

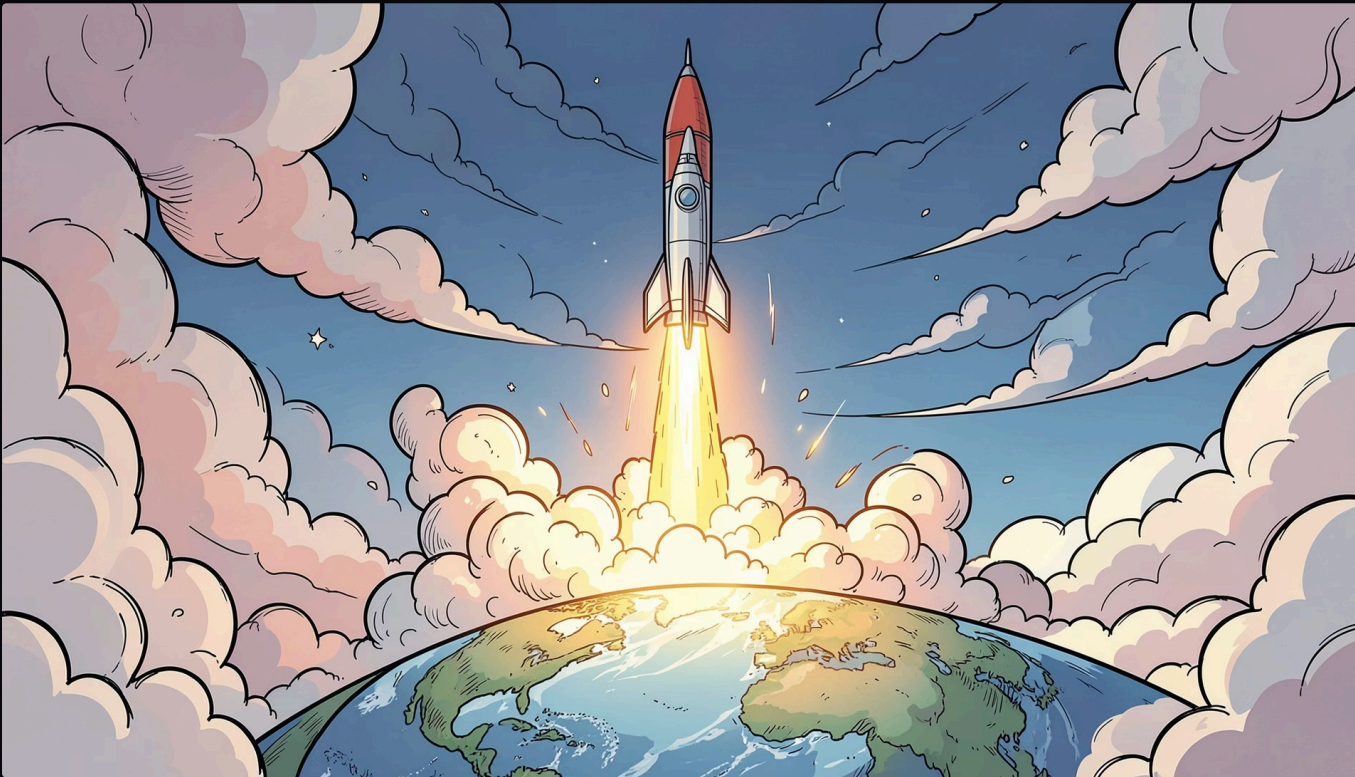
$$g = \frac{GM}{R^2} \approx 9.8 \text{ m/s}^2$$

Variation of g

- **With altitude (h):** $g_h = g \left(1 - \frac{2h}{R}\right) \rightarrow g$ **decreases**
- **With depth (d):** $g_d = g \left(1 - \frac{d}{R}\right) \rightarrow g$ **decreases**
- **At Earth's center:** $g = 0$
- **With latitude:** g is **maximum at poles**, minimum at equator (due to rotation)



Gravitational Potential Energy



Definition

Work done in bringing a body from infinity to a point in the gravitational field without acceleration.

Formula

$$U = -\frac{GMm}{r}$$

- U is always **negative** (attractive force)
- At infinity, U = **0**
- At Earth's surface: $U = -\frac{GMm}{R}$

□ **Gravitational Potential (V):**
Work done per unit mass.

$$V = -\frac{GM}{r}$$

Escape Velocity

Definition

The minimum velocity required for a body to escape from the gravitational field of a planet and never return.

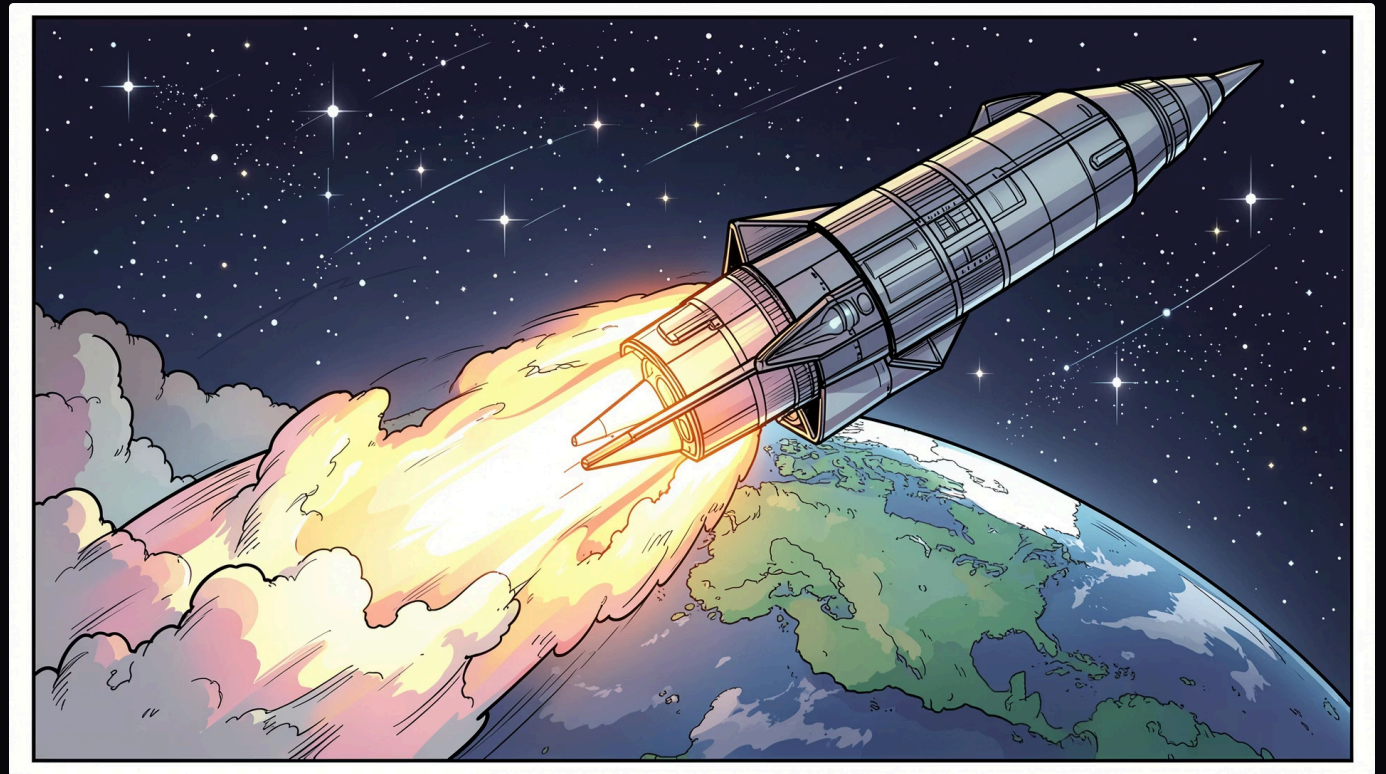
Formula

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR}$$

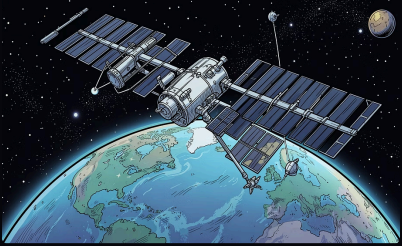
For Earth

Escape velocity = 11.2 km/s

- Independent of the **mass** of the body
- Depends only on mass and radius of the planet
- Direction of projection does **not** matter

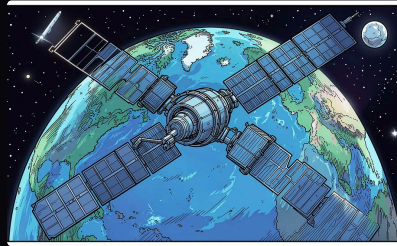


Satellites — Types & Orbital Velocity



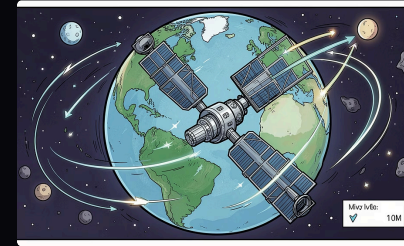
Geostationary Satellite

Orbits in equatorial plane, period = **24 hours**, height $\approx$ **36,000 km**. Used for TV & communication.



Polar Satellite

Orbits over poles, low altitude (500–800 km). Used for **weather forecasting** and remote sensing.



Orbital Velocity

$v_o = \sqrt{\frac{GM}{r}}$. For near-Earth orbit: $\approx$ **7.9 km/s**. Independent of satellite's mass.

Time Period & Energy of a Satellite

Time Period (T)

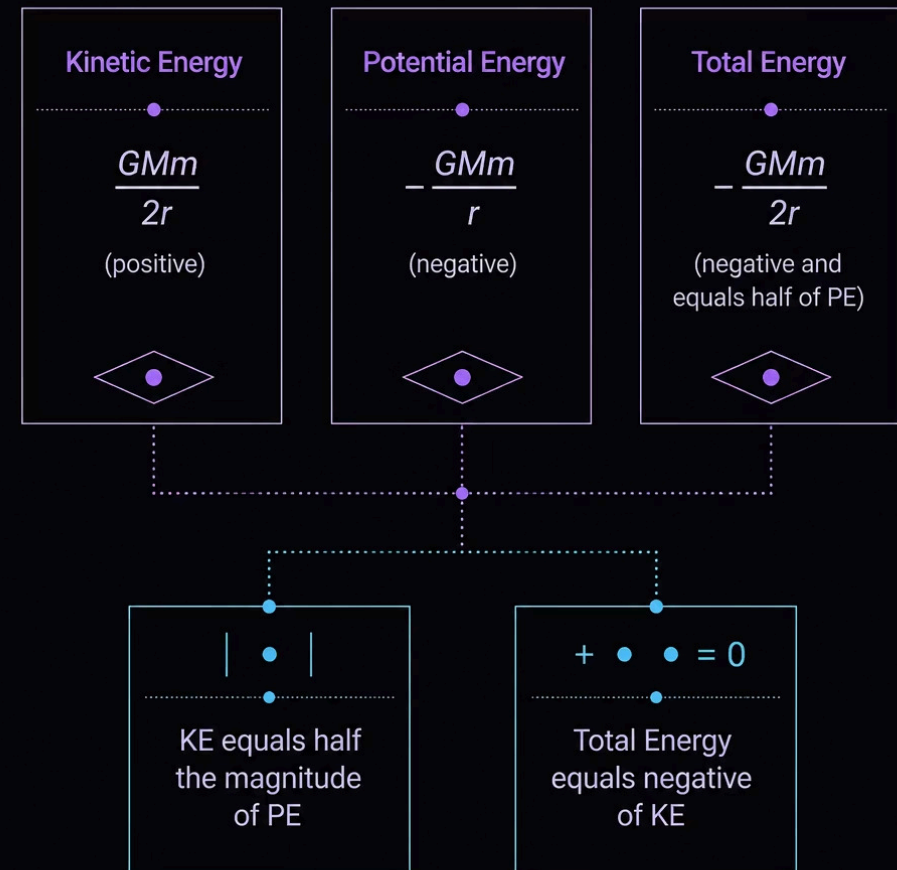
$$T = 2\pi\sqrt{\frac{r^3}{GM}}$$

For a satellite close to Earth's surface: **T ≈ 84.6 minutes**

Total Energy of Satellite

$$E = K.E. + P.E. = -\frac{GMm}{2r}$$

- Total energy is **negative** → satellite is bound
- $K.E. = \frac{GMm}{2r}$
- $P.E. = -\frac{GMm}{r}$
- $|E| = K.E. = \frac{|P.E. |}{2}$



Key Formulas at a Glance

Concept	Formula	Value / Note
Gravitational Force	$F = GMm/r^2$	$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
Acceleration due to gravity	$g = GM/R^2$	9.8 m/s ² on Earth's surface
Gravitational Potential Energy	$U = -GMm/r$	Always negative
Escape Velocity	$v_e = \sqrt{(2GM/R)}$	11.2 km/s for Earth
Orbital Velocity	$v_o = \sqrt{(GM/r)}$	$\approx 7.9 \text{ km/s}$ (near Earth)
Time Period	$T = 2\pi\sqrt{(r^3/GM)}$	$\approx 84.6 \text{ min}$ (near Earth)
Kepler's 3rd Law	$T^2 \propto a^3$	Valid for all planets

Chapter 8 — Key Takeaways

→ Universal Gravitation

Every mass attracts every other mass — force follows inverse square law.

→ Kepler's Laws

Describe planetary motion — elliptical orbits, equal areas, $T^2 \propto a^3$.

→ g Varies

Decreases with altitude, depth, and from poles to equator.

→ Escape & Orbit

Escape velocity = 11.2 km/s; Orbital velocity = 7.9 km/s near Earth.

